

Nano Bacuri Açaí

Luís Romera

In this document I'll be analyzing the interaction between **Bacuri**, **Açaí** and **Brij O10** in the formulation of **nanoparticles**.

So first I'll import the data:

```
setwd(
  "C:/Users/luisr/Documentos/Documents/Documentos - Copia/NanoIA/Topicos_em_Nanotecnologia/NanoBacuriAcai/"
)

data <- read.csv(
  "NanoBacuriAcai.csv"
)

head(
  data
)
```

	Bacuri	Acai	Brijo10	PDI	Size
1	0.2500	0.2500	0.5000	0.163	737.00
2	0.1665	0.5000	0.3335	0.178	68.00
3	0.5000	0.1665	0.3335	1.000	590.00
4	0.3750	0.3750	0.2500	0.691	504.00
5	0.2500	0.5000	0.2500	0.183	223.00
6	0.2500	0.1665	0.5835	0.689	495.96

```
View(
  data
)
```

Now, I'll adjust a model:

```
library(
  mixexp
)

mixvars <- c("Bacuri", "Acai", "Brijo10")

lin <- MixModel(
  data,
  "PDI",
  mixcomps = mixvars,
  1
)
```

	coefficients	Std.err	t.value	Prob
Bacuri	2.3444362	0.3393683	6.9082363	0.0001234984
Acai	-0.3494541	0.2925435	-1.1945373	0.2664794382
Brijo10	-0.1542760	0.2489946	-0.6195955	0.5527573264

Residual standard error: 0.1759035 on 8 degrees of freedom
 Corrected Multiple R-squared: 0.7779811

Let's begin the model diagnostic:

```
summary(lin)
```

Call:

```
lm(formula = mixmodnI, data = frame)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.25933	-0.05386	-0.01884	0.03679	0.26824

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
Bacuri	2.3411	0.3402	6.881	0.000127 ***
Acai	-0.3464	0.2933	-1.181	0.271561
Brijo10	-0.1527	0.2498	-0.611	0.557991

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1765 on 8 degrees of freedom
 (4 observations deleted due to missingness)
 Multiple R-squared: 0.9469, Adjusted R-squared: 0.927
 F-statistic: 47.53 on 3 and 8 DF, p-value: 1.917e-05

```
anova(lin)
```

Analysis of Variance Table

Response: PDI

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Bacuri	1	4.3698	4.3698	140.2735	2.369e-06 ***
Acai	1	0.0608	0.0608	1.9530	0.1998
Brijo10	1	0.0116	0.0116	0.3736	0.5580
Residuals	8	0.2492	0.0312		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Let's now adjust a quadratic model

```
quad <- MixModel(  
  data,  
  "PDI",  
  mixcomps = mixvars,  
  2  
)
```

	coefficients	Std.err	t.value	Prob
Bacuri	2.626767	1.926203	1.3637026	0.2308471
Acai	1.770859	1.814958	0.9757023	0.3740229
Brijo10	1.836452	1.294917	1.4182004	0.2153368
Acai:Bacuri	-1.941341	5.651068	-0.3435353	0.7451700
Brijo10:Bacuri	-3.033517	5.654223	-0.5365047	0.6146111
Acai:Brijo10	-9.380093	5.023156	-1.8673706	0.1208210

Residual standard error: 0.1624006 on 5 degrees of freedom
 Corrected Multiple R-squared: 0.8817241

```
anova(quad)
```

Analysis of Variance Table

Response: PDI

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Bacuri	1	4.3698	4.3698	165.9243	5.022e-05 ***
Acai	1	0.0608	0.0608	2.3101	0.1890
Brijo10	1	0.0116	0.0116	0.4420	0.5356
Bacuri:Acai	1	0.0242	0.0242	0.9192	0.3817
Bacuri:Brijo10	1	0.0004	0.0004	0.0156	0.9055
Acai:Brijo10	1	0.0929	0.0929	3.5282	0.1191
Residuals	5	0.1317	0.0263		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Let's compare the two models:

```
anova(lin, quad)
```

Analysis of Variance Table

Model 1: PDI ~ -1 + Bacuri + Acai + Brijo10

Model 2: PDI ~ -1 + Bacuri + Acai + Brijo10 + Bacuri:Acai + Bacuri:Brijo10 +
Acai:Brijo10

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	8	0.24922				
2	5	0.13168	3	0.11754	1.4876	0.3249

Conclusion

We saw that the quadratic model doesn't differ significantly from the linear model. So we may hypothesize, based on our data, that the relation between the components is linear. Also, Bacuri was the only parameter that significantly shapes the response.